Market-State-Conditioned Orchestration of Heterogeneous Agents for Financial Decision Support: A Colombo Stock Exchange Study

Investigating how the influence of methodologically heterogeneous financial agents should be set when the reliability of their evidence channels varies with market conditions, by designing, implementing and evaluating a market-state-conditioned orchestration operator on Colombo Stock Exchange data.

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Academic Year: 2026–2027 (Group 4)

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THE DUAL-SPINE INQUIRY STRUCTURE

RESEARCH SPINE

Problem Context: Agent reliability varies with market state (Slide 3).

The Void: Clause-by-clause comparative gap matrix (Slide 4).

Inquiry: Pre-registered research questions and proof methods (Slide 5).

Proof Engine: Five-arm evaluation over cached outputs (Slide 10-11).

SOFTWARE ENGINEERING SPINE

Need: Prevent confident output from an empty information channel (Slide 9).

Mechanism: Topological architecture of the orchestration operator (Slide 6-7).

Feasibility: Reconciled data groundwork and boundaries (Slide 8-9).

Guardrails: Traceability, risk diagnostics, and QA (Slide 12-14).

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AGENTS PRODUCE CONFIDENT OUTPUT ABOUT NOTHING WHEN CHANNELS GO THIN

Pavithya et al., 2021

Fixed Dials. Two models on the CSE combined at a fixed 3:1 ratio. Realized influence drifted entirely based on article volume, completely unmanaged by the architecture.

AlphaAgents

Manual Intervention. An entire agent had to be deleted by hand because its input channel went too megacaps. The system had no abstention mechanism.

FinAgent

Static Tuning. Auxiliary strategies returned 16% against buy-and- 29% because they were tuned once to conditions that subsequently changed.

THE CSE STRESS-TEST CONTEXT

The Colombo Stock Exchange is utilized as a frontier-market stress-test environment where channel reliability varies dramatically.

- Mean institutional ownership 66.1% (vs 5.7% insider) meaning available depth is worse than capitalization suggests.
- Unexpected illiquidity shocks depress returns at −0.084 to −0.097.

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THE CLAUSE-BY-CLAUSE ARCHITECTURAL VOID

	CONTINUOUS WEIGHTS	INFLUENCE (NOT SELECTION)	METHODOLOGICAL HETEROGENEITY	SINGLE SHARED DECISION	MEASURED EXOGENOUS STATE
MacroHFT	Yes	Yes	**NO** (6 identical networks)	Yes	Yes
TRIAG	Yes	Yes	Yes	Yes	**NO** (State vector lacks environment block)
HedgeAgents	Yes	**NO** (Weights assets, not methods)	Yes	**NO** (One agent per asset)	Yes
Pavithya et al.	**NO** (Fixed at design)	Yes	Yes	Yes	**NO**
The Proposed Target	✓	✓	✓	✓	✓

LINEAGE STATEMENT: A generalization of feature-conditioned forecast combination (FFORMA) along an identified axis: conditioning on measured, exogenous market state at each step, rather than static properties of the series.

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PRE-REGISTERED INQUIRY AND PROOF METHODS

OVERARCHING GOAL

Establish under what market conditions market-state-conditioned orchestration earns its place against fixed, static learned, and performance-reactive weighting.

MAIN RESEARCH QUESTION (MRQ)

How should the influence of heterogeneous agents be adapted to measured market state?

Proof Method: Five-arm evaluation over byte-identical cached outputs with per-input ablations

SQ1: LIQUIDITY & EXECUTION

How should a system alter execution behavior and price-derived agent influence when liquidity deteriorates?

Proof Method: The A3 sensitivity sweep (∂w / ∂L) and separating ablations

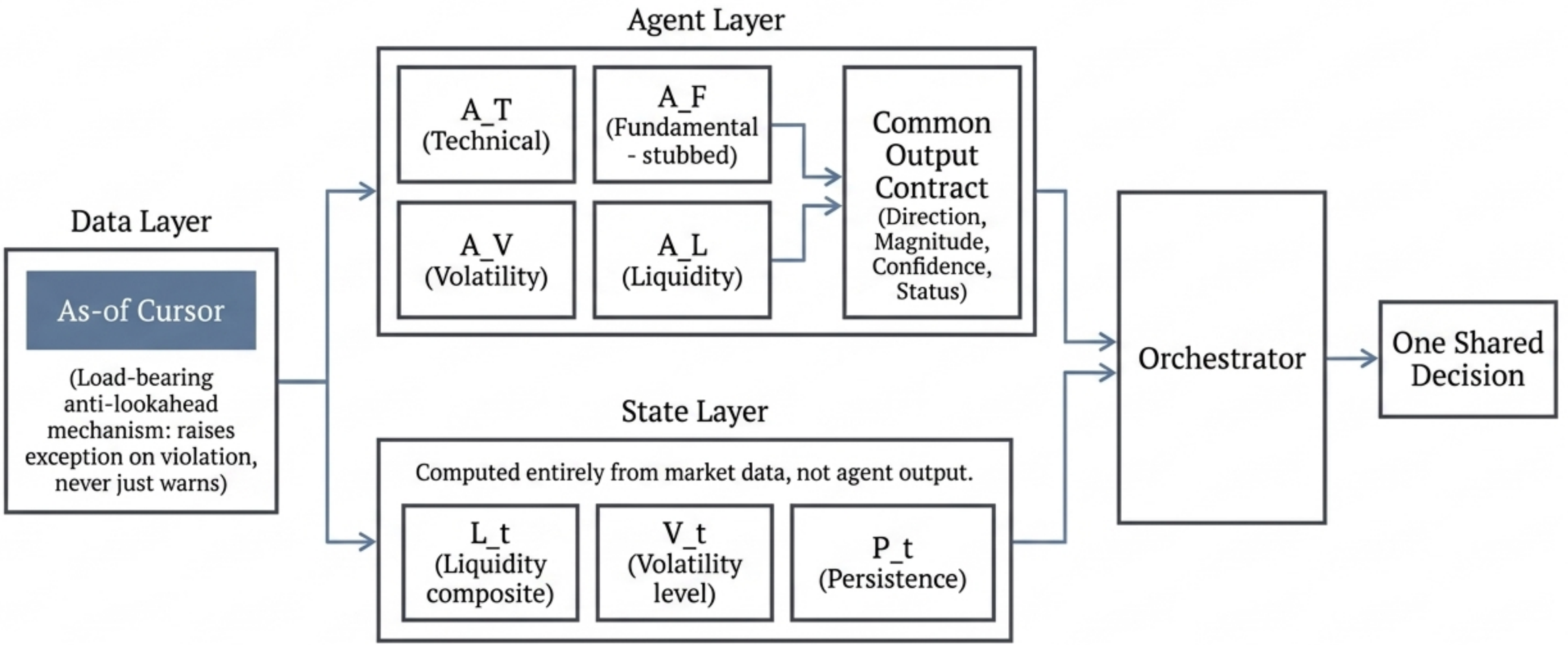
SQ2: COMPARABILITY

How can confidence of unlike architectures be placed on a common scale?

Proof Method: Four structural confidence mechanisms mapped through rolling empirical accuracy-at-threshold curves

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TOPOLOGICAL SEPARATION OF AGENT OUTPUT AND MARKET STATE



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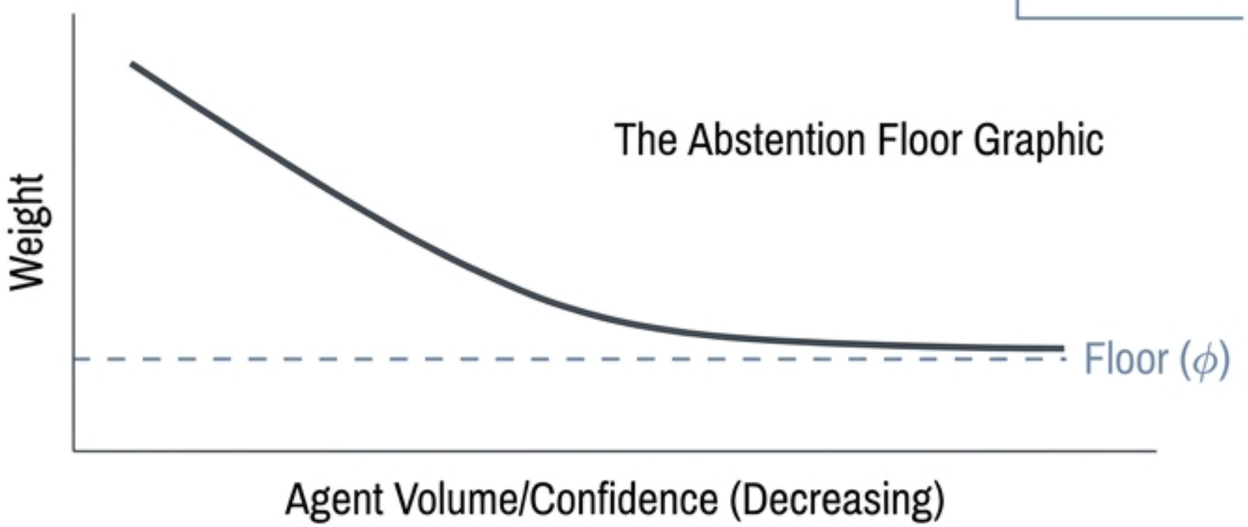
THE ORCHESTRATION OPERATOR AND THE ABSTENTION FLOOR

Linked to CSE evidence that trading volume Granger-causes the index at $p = 0.0041$.

Linked to local long-memory feasibility confirmed at $\hat{d} = 0.408$ on absolute returns.

$$w_{i,t} = f(L_t, V_t, P_t, C_{i,t})$$

Structural confidence, specifically not self-reported by agents.



The Abstention Floor Graphic

Prevents losing the channel while protecting from false confidence.

LOW-CAPACITY FORM

Deliberately chosen as a gated linear form over a neural network to prevent entropy collapse, as observed in nearest-neighbor architectures.

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DATA FOUNDATION AND RECONCILIATION SECURED

99.93%

Archive agreement across 817,000 rows.

0.13%

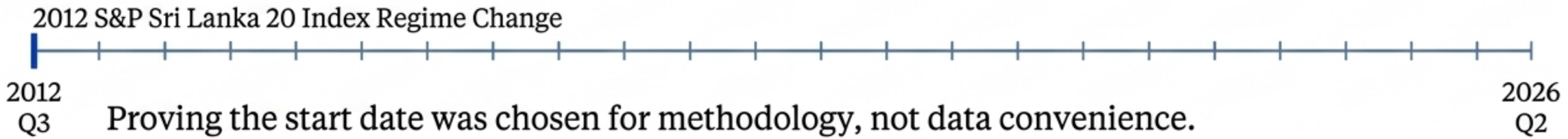
Turnover reconciliation against published annual figures.

3,920

Validated trading days matching CSE annual reports (17 years).

22,893

Filings secured for the fundamental agent, every filename prefixed with the strict announcement date.



Proving the start date was chosen for methodology, not data convenience.

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ENGINEERING CONSTRAINTS AND DELIBERATE EXCLUSIONS

EXEMPLAR REQUIREMENTS

FR1 (As-of Cursor): Point-in-time correctness. Cursor violation raises an exception, never a warning. (Prevents published CSE backfill failures).

NFR3 (Reproducibility): Byte-identical cached arm isolation.

DR2 (Corporate Actions): Tr_close is primary against ASTRI. 31 register corrections applied.

MAI1 (Structural Confidence): Prevents agents from hallucinating self-assessed certainty.

STRICT BOUNDARIES (EXCLUSIONS)

No Sentiment Analysis: Final decision by supervisor; avoids a second research problem.

No Microstructure Execution: Proxy route established. CSE does not supply bid/ask/depth in the 40-item library.

No 5th Agent (Macro): Rejected on evidence (ARLD gave adj $R^2 = 0.189$ over a decade with Granger nulls).

No UI/Front-End: Bounded strictly as a batch research pipeline.

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EXPERIMENTAL ARMS AND THREAT ISOLATION

B_1 (Equal Weight)	The field's hardest benchmark. Tests if weighting is worth the estimation error.
B_2 (Static Learned)	Tests if fixed learned weights achieve the same result without continuous conditioning.
B_3 (Performance-Reactive)	The primary threat. Tests if merely reacting to recent agent performance does just as well as exogenous market state conditioning.
P (Proposal)	State-conditioned orchestration.
O (Oracle)	Fitted on the test window. An unattainable diagnostic upper-bound to price estimation cost in local units, separating a failed hypothesis from a hard estimation problem.

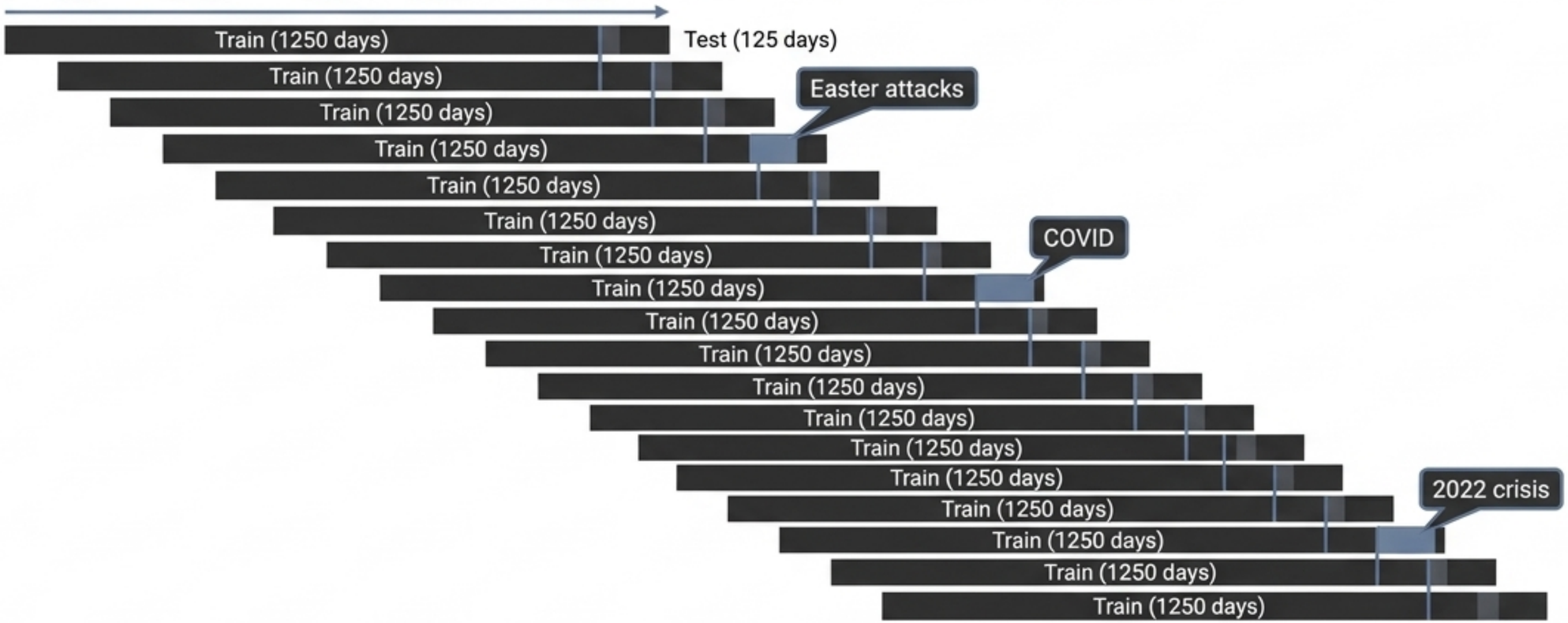
Fifty years of evidence shows Equal Weighting is persistently hard to beat. It is the pre-registered primary benchmark. A null result against it is an acceptable and reportable outcome.

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PRE-REGISTERED WALK-FORWARD PROTOCOL AND SUCCESS CRITERION

Margin notes:

Periodic refit at each boundary. Crises fall deliberately inside test blocks to ensure regime variation.



THE MARGIN

$$\text{median}(\Delta_k) > \text{IQR}(R_{B_1,k})$$

Primary Metric: Cost-adjusted return over the full window, abstained days held flat.

Explanation: Improvement over B_1 must exceed the baseline's own fold-to-fold dispersion.

Paired with HAC and stationary bootstrap significance ($\alpha = 0.05$).

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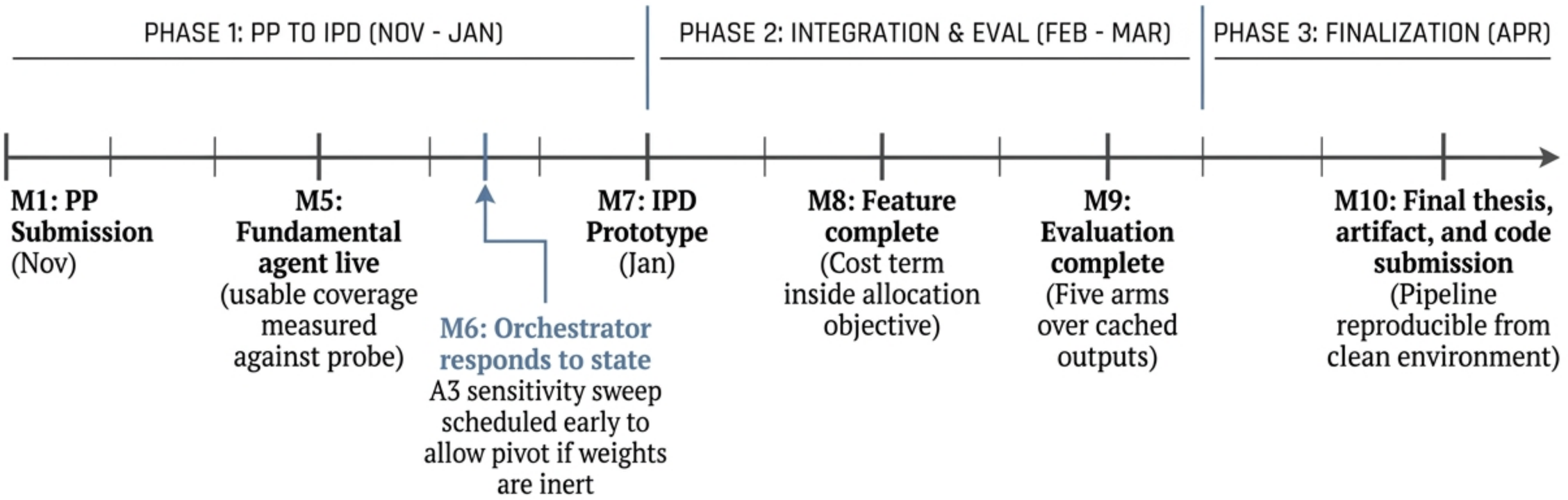
SYSTEM INTEGRITY AND TRACEABILITY ASSESSMENTS

Test Name	What it asserts
Cursor Violation	Query past as-of cursor strictly raises CursorViolation; no bypass permitted.
Arm Isolation	Output files must resolve byte-identical across all five evaluation arms to ensure weighting is the only variable.
Simplex Invariant	Weights must sum to 1.0; no non-unparseable agent falls below floor ϕ.
Calibration Monotonicity	Accuracy-at-threshold transforms must be strictly monotone and bounded in [0,1].
Decision Reconstruction	A randomly sampled decision must be 100% reconstructable from the per-ticker-date log alone (proving K7 contribution is verifiable).

WEIGHT ENTROPY CHECK
Evaluates $H_t = -\sum w_{i,t} \ln w_{i,t}$ to detect orchestration collapse (where weights freeze near-constant over over time).

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DELIVERY SCHEDULE AND CRITICAL PATH TO APRIL 2027



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RISK DIAGNOSTIC AND PRE-DECLARED FALLBACKS

Risk Domain	Likelihood & Impact	Mitigation & Fallback
Research Risks		
State conditioning adds nothing beyond recent performance	Moderate / High	Mitigation: B_3 arm isolates this exactly; reportable as a null.
Entropy collapse (near-constant weights)	Moderate / High	Mitigation: Diagnostic pre-registered; if observed, trace is reported.
Technical & Data Risks		
~~Price series not adjusted for corporate actions.~~	Closed	Adjusted panel built Oct 23 (adj_close / tr_close validated).
Bid, ask, and depth data never arrive from CSE	High / Moderate	Fallback: Level A proxy route (illiquidity from volume, turnover, Amihud) is declared and guaranteed feasible.
Fundamental agent unbuildable in time	Moderate / Moderate	Fallback: Stubbed against probe set while emitting contract; on-and-off ablation runs in degenerate sense, reported honestly.

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CLOSING DEFENSE AND GENERALIZATION

THIS PROJECT ADDRESSES...

How the influence of methodologically heterogeneous agents should be dynamically set over a single shared decision when their evidence channels' reliability fluctuates.

IT IS NEEDED BECAUSE...

Existing systems fix this influence at design time (or lack a weight object entirely), causing architectures to remain blind to documented channel collapses on frontier markets.

THE PROPOSED CONTRIBUTION IS...

A continuous, state-conditioned orchestration operator (generalizing FFORMA), paired with attributable evidence on whether it earns its place against place against reactive weighting.

THE SCOPE IS CREDIBLE BECAUSE...

Boundaries are fixed, the hardest baseline (B_1) is pre-registered as primary, a null outcome is explicitly acceptable, and data infrastructure is already built and reconciled.

SCOPE BOUNDING

Bounded strictly to one market, simulated execution, and no sentiment. An answer here is structurally valuable to the field even if the answer is that equal weights win.

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